Part c.

In steady state, $\ln(z_t^i) = \ln(z_{t-1}^i) = \ln(z^i)$, $\varepsilon = 0$.

Therefore,

$$US: \ln(z^{us}) = \chi^{us} + \rho^{1,us}\ln(z^{us}) + \phi^{1,us}\ln(z^{uk})$$
$$Uk: \ln(z^{uk}) = \chi^{uk} + \rho^{1,uk}\ln(z^{uk}) + \phi^{1,uk}\ln(z^{us})$$

$$\ln(z^{us})(1-\rho^{1,us}) = \chi^{us} + \phi^{1,us}\ln(z^{uk})$$

$$\ln(z^{us}) = \frac{\chi^{us} + \phi^{1,us}\ln(z^{uk})}{1-\rho^{1,us}}$$

Substitute it into Uk. function.

$$\ln(z^{uk})(1-\rho^{1,uk}) = \chi^{uk} + \phi^{1,uk}\left[\frac{\chi^{us} + \phi^{1,us}\ln(z^{uk})}{1-\rho^{1,us}}\right]$$

$$\ln(z^{uk}) = \frac{\chi^{uk}(1-\rho^{1,us}) + \phi^{1,uk}\chi^{us}}{(1-\rho^{1,uk})(1-\rho^{1,us}) - \phi^{1,uk}\phi^{1,us}}$$

We have $\chi^{us} = 0.018939$, $\chi^{uk} = 0.011015$

$\rho^{1,us} = 1.001897$, $\rho^{1,uk} = 0.768823$

$\phi^{1,us} = -0.041205$, $\phi^{1,uk} = 0.156165$

$$\ln(z^{uk}) = \frac{0.011015\cdot(1-1.001897) + 0.156165\times 0.018939}{(1-0.768823)(1-1.001897) - 0.156165\times(-0.041205)}$$

$$= \frac{-0.0000209 + 0.0029576}{0.231177\times(-0.001897) + 0.006435}$$

$$= \frac{0.002450\,1}{0.0059965} = 0.489736$$

$$\ln(z^{us}) = \frac{0.018939 + (-0.041205) \times 0.489736}{(1 - 1.001897)}$$

$$= \frac{-0.000124}{-0.001897} = 0.653664$$

$$z^{us} = e^{0.653664} = 1.923572$$

$$z^{uk} = e^{0.489736} = 1.631919$$